Solutions to Machine Learning WORKSHEET-1

1. Least Square Error
2. Linear regression is sensitive to outliers
3. Negative
4. Correlation
5. None of these
6. Predictive model
7. Regularization
8. SMOTE
9. Sensitivity and Specificity
10. False
11. Apply PCA to project high dimensional data
12. It becomes slow when number of features is very large and It does not make use of dependent variable.
13. Regularization refers to techniques that are used to calibrate machine learning models in order to minimize the adjusted loss function and prevent overfitting or underfitting. Using Regularization, we can fit our machine learning model appropriately on a given test set and hence reduce the errors in it. Regularization is a technique in machine learning which is used when multicollinearity exists in your dataset and also when you wish to perform variable selection (reduce the number of predictors in your model). Regularization is used to reduce the variance in your model at the cost of introducing some bias.
14. There are 3 types of regularization - Ridge Regression, LASSO (Least Absolute Shrinkage and Selection Operator) Regression, and Elastic Net. The main difference between Lasso and Ridge is the penalty term they use. Ridge uses the L2 penalty term which shrinks the coefficients towards zero, and limits the size of the coefficient vector. Ridge regression is used to reduce multicollinearity in your dataset. It cannot be used for variable selection. LASSO uses the L1 penalty, which imposes sparsity among the coefficients and reduces some of the coefficients to exactly 0, thus performing variable

selection and making the fitted model more interpretable. LASSO does not have a closed form solution because the objective function is not differentiable. Elastic Net leverages the benefits of both Ridge and LASSO, and has a penalty which is a mix of $L1$ and $L2$ norms. Glmnet is an effective R package that I've used to implement regularization. Ridge(), Lasso(), and LassoCV() are some useful functions in the scikit-learn library in Python, which we can use to implement L2 and L1 regularization. It is recommended to scale your input features before regularization.

15. The error term is the stuff that isn't explained by the model. For a very simple example, suppose you are predicting the weight of adult human males based on their height. Well, height is certainly related to weight - taller people tend to be heavier - but the model won't be perfect because there is a range of weights at each height. The error is the difference between the predicted value and the actual value.